

Empirical Li-ion aging model derived from single particle model

Sabrina Kathrin Rechkemmer^{a,b,c,*}, Xiaoyun Zang^b, Weimin Zhang^c, Oliver Sawodny^a

^a Institute for System Dynamics, University of Stuttgart, Stuttgart 70563, Germany

^b Bosch (China) Investment Ltd., 200335 Shanghai, China

^c School of Mechanical Engineering, Tongji University, 201804 Shanghai, China

ARTICLE INFO

Keywords:

Li-ion
Electrical equivalent model
Empirical aging model
Single particle model
LMO
Aging derivation

ABSTRACT

Battery lifetime simulation and optimization still suffers from the need of accurate but not too computationally intensive models. Electrical equivalent models (EEMs) are often used in automotive industry due to their efficiency. However, aging extensions for EEMs are typically based on empirical relations and fail outside their design scope. Single particle models (SPMs) on the other hand, are one of the simplest forms of electrochemical models which might still be suitable for automotive applications. Although EEM and SPM are differently motivated and developed, cross connections between both model types are discussed. Based on a simplification of SPM aging relations, a more comprehensive empirical aging model for EEMs denoted by SPM–EEM is described and compared to already existing models. As an example, both models are adjusted to fit a LiMn_2O_4 (LMO) cell chemistry. The derived SPM–EEM shows first promising results to improve the accuracy. Even though SPMs are slightly superior in predictability than EEMs, they are often too complex to run on control devices. Improving the predictability of EEMs is, therefore, an important step to improve aging estimation and optimization for automotive applications.

1. Introduction

Nowadays, electric vehicles are promoted as green alternative to traditional combustion cars. This is especially true for China, as the cities suffer from severe air pollution [1,2]. Nevertheless, most electric vehicles still cannot live up to the users expectations for range and lifetime. In most cases, battery technology is the bottleneck to increase customer acceptance. Driving range with one battery charge as well as battery recharging times are still not competitive to conventional vehicles. Additionally, Li-ion batteries are still quite expensive and in most applications reach their end of life before their expected lifetime [3,4]. Battery lifetime can be increased by limiting the operation range and current. Accurate and easy to compute Li-ion battery aging models are still rare. Simple models are desired to optimally design the operation range for each case and environmental condition.

The tradeoff between accuracy and computation time for optimization purposes leads to the need of simple models, which still cover all relevant effects. For automotive purposes, electrical equivalent circuits are commonly used as described in [5–7]. For simulation and design purposes on the other hand, more accurate electrochemical models, which account for internal cell processes, are applied [8].

There is a growing body on Li-ion battery and cell models. On the one hand physical models such as electrical equivalent models (EEMs),

which neglect the causes of capacity and resistance changes and mainly rebuild the voltage and current characteristics, are available. On the other hand electrochemical models of various modeling depth try to resemble cell internal processes including transport of ions and chemical reactions. Inside electrochemical models, molecular models such as Kinetic Monte Carlo and multiphysics models offer the highest accuracy but require also the highest computation power and system knowledge. Pseudo two dimensional models use simplified expressions of the electrodes and reactions [9]. Single particle models (SPMs) [10,11] are one dimensional models which neglect different concentrations in the solution and merely model the electrodes as one particle where diffusion and intercalation occurs. For a detailed comparison it is referred to [8]. Various approaches have been presented to further simplify SPMs by converting the partial differential equations (PDEs) to ordinary differential equations (ODEs). They are based on Pade approximation [12], orthogonal decomposition [13], and many other simplification methods [9,14]. However, they lose the physical properties of the model parameters. The aim of this paper is, therefore, not to simplify the SPM by model approximation methods but to incorporate the model knowledge of the SPM aging relations into the aging model of an EEM. An improved SPM–EEM ampere counting aging model is derived by simplifying the SPM aging equations and describing them by measures which are also available for an EEM such as voltage

* Corresponding author at: Institute for System Dynamics, University of Stuttgart, Stuttgart 70563, Germany.

E-mail address: sabrina.rechkemmer@isys.uni-stuttgart.de (S.K. Rechkemmer).

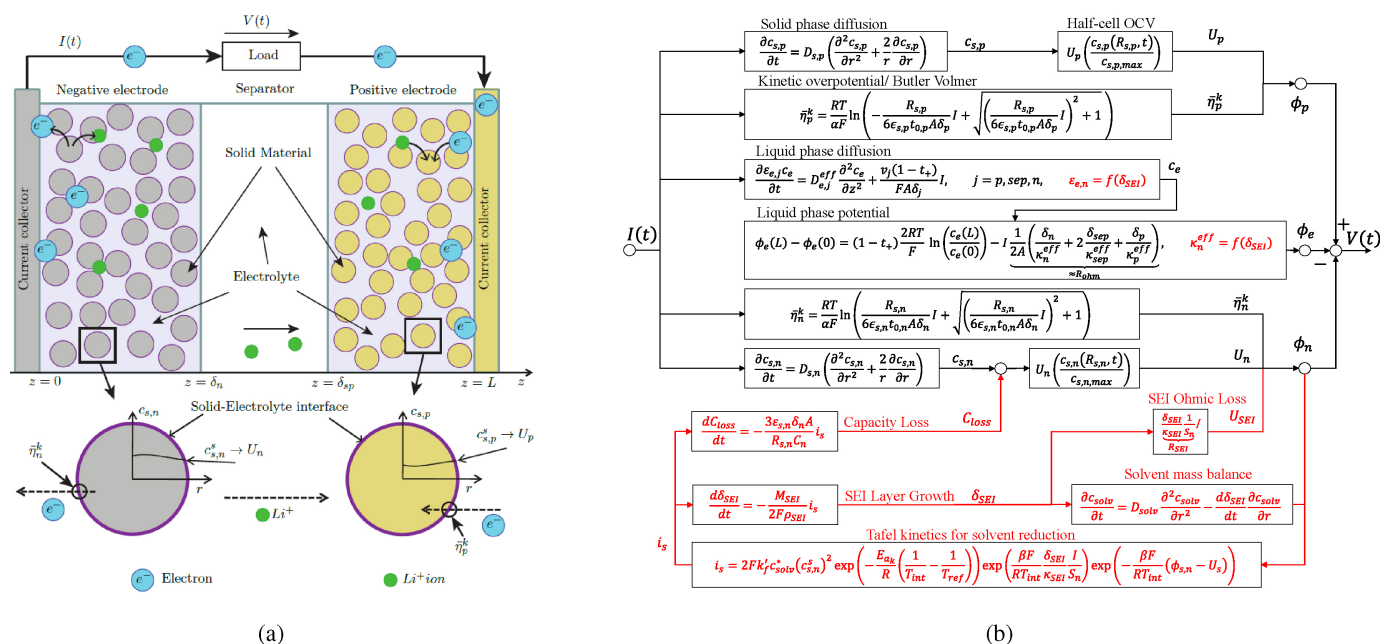


Fig. 1. (a) Derivation of a single particle model (SPM) [31,32] and (b) basic electrochemical equations (top, black) including aging extension (bottom, marked in red). For a detailed description of all parameters, refer to [Appendix C](#). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and temperature instead of concentrations.

At first, a short introduction to aging processes and state of art empirical aging modeling is provided. The basics of SPM modeling are introduced in Section 3 focusing on the description of aging effects. Section 4 infers the improved SPM-EEM aging equation from the electrochemically motivated SPM aging description. The SPM-EEM using the derived empirical aging equation is in Section 5 compared to an existing EEM relation presented by [5]. Section 6 provides an overview on how to adapt the models to a different cell chemistry based on literature data and aging measurements to motivate the lower complexity of an EEM in contrast to a SPM. Conclusions are drawn in Section 7.

2. Aging modeling

Aging effects are directly represented in the equations of electrochemical models. They mainly differ in their extent and depth of modeling different aging processes. Aging processes result from different chemical processes, transport processes, external stress, and their interactions. Although mainly interactions on the electrodes' surface and electrolyte lead to aging effects, also current collectors, binders and additives might interact [15]. Aging processes can be categorized into loss of active material (LAM), for example due to volume changes of graphite, and loss of lithium inventory (LLI). LLI results from side reactions such as electrolyte decomposition, formation of a passivation layer, and lithium plating. LAM mainly occurs due to mechanical stress-induced cracking and dissolution during operation. For example, volume and structural changes occur. Refer to [16–19,15] for details. Aging mechanisms are commonly separated for negative and positive electrode although similar effects occur.

Main reason for capacity loss and resistance increase is accounted to the growth of the so-called solid electrolyte interface layer (SEI) at the negative electrode (graphite) [20–22]. A SEI layer forms as the graphite at the negative electrode's surface reacts with electrolyte components due to a negative potential being applied. During the first cycles, this SEI layer is built up preventing further side products to form. It is commonly assumed to be separated into a smaller more stable inorganic layer and a porous organic layer consisting of various side products

formed by the salts and solvents in the electrolyte as well as Li-ions [20–22]. Ideally, Li-ions can pass through the SEI layer and electrons cannot by being immediately consumed. However, due to electrical and mechanical stress during charge and discharge, the SEI layer can be damaged. During repair processes, the SEI layer consumes further Li-ions. Occurring processes are, for example, summarized in [23] as electrolyte decomposition leading to the formation of a SEI, graphite exfoliation due to cracks formed by gas and solvent co-intercalation, SEI growth and porosity changes, corrosion effects at the current collector, and lithium plating. These processes can be further enhanced due to ions formed by the positive electrode material (e.g. Mn^{2+}). Especially at high or low temperatures these effects become apparent and are enhanced by non-uniform temperature distributions within the cell [24,25]. In total, the SEI layer grows and leads to a capacity loss and resistance increase as Li-ions are constantly consumed and the SEI thickness adds to the cell's electrode resistance. Once the cell is close to its end of life, which is commonly defined after reaching 80% of the cell's initial capacity or a doubling in resistance, the aging characteristic changes. A rapid loss in capacity and resistance increase is observed. Whereas the growth of the SEI is predominantly responsible for aging before the end of life criteria is reached, lithium plating accounts for the rapid degradation after end of life [26,27].

Due to the manifold lithium metal compounds used as positive electrode material, it is hard to formulate occurring reactions and interactions exactly. Similar to the negative electrode, the active material (lithium metal oxide) changes, for example due to structural changes (phase transitions). The positive electrode comprising lithium also reacts with the electrolyte forming a SEI, but this effect is negligible compared to the passivation effect of the SEI on the graphite surface. Further, electric and mechanic connections between current collector, binder, and active material might loosen, and LAM occurs as Li-ions might interact with the negative electrode [23,19].

Li-ions are formed at the lithium electrode and move from it to the graphite electrode due to the potential difference. They migrate through the SEI at the graphite electrode and interact with the graphite. Fig. 1(a) illustrates this process. Although various interactions and reactions occur during this process, it is often assumed that the migration of Li-ions through the SEI is the rate-determining process and can be

modeled to represent the whole reaction kinetics as described in [28,21,22,29,30]. Simplified, SEI layer thickness δ_{SEI} increases based on its molar mass, density, and the reaction rate i_s of the dominant side reaction. The reaction rate i_s in turn depends on the solvent concentration difference between the surface of the negative electrode and the surface of the SEI. Depending on the modeling depth of the electrochemical model, these relations are represented on a three dimensional molecular, two dimensional particle or even one dimensional single particle scale. Due to the wide variety of electrochemical models, aging equations are not discussed in detail in this study. A SPM implementation is provided in Section 3.

For EEM on the other hand, various empirical equations have been derived in the past years. However, they show many similarities. In the following, focus is laid on ampere-counting empirical equations. As they require little background information and result in simple expressions, they are commonly used in applications where low computation power is available such as battery management systems, or where calculations have to be repeated such as in optimizations.

One of the first ampere-counting empirical aging equations for Li-ion batteries was published in [33]. Bloom et al. used a temperature T and time t dependent power law to represent calendar aging as shown in (1). In (1), A is a parameter with no direct physical representation, E_a the activation energy, R the gas constant and z the power factor, which was set to 0.5 as it was thought to represent the SEI layer growth. This equation was adapted to a LiFePO cell chemistry by [34]. Wang et al. extended the equation to cycle aging by replacing time with the total charge throughput Q_{Ah} as seen in (2).

$$C_{loss} = A \exp\left(\frac{-E_a}{RT}\right) t^z \quad (1)$$

$$C_{loss} = A \exp\left(\frac{-E_a}{RT}\right) (Q_{Ah})^z \quad (2)$$

These ampere-counting equations in (3) are expanded by including voltage V and depth of discharge (DOD) dependencies which are, for example, described in [5].

$$C_{loss} = a_1(V + a_2) \exp\left(\frac{-E_a}{RT}\right) t^{z_1} + (b_1(\emptyset V + b_2)^2 + b_3 DOD + b_4) Q_{Ah}^{z_2} \quad (3)$$

Calendar and cycle aging are integrated in one aging equation. Other approaches switch between calendar and cycle aging, using two separate equations. In case of calendar aging [10] introduced Eq. (4) and for cycle aging equation (5). Once capacity degradation is modeled by a state of charge (SOC) dependent parameter A . The same parameter is a function of the current I for cycle aging.

$$C_{loss,cal} = A(SOC) \exp\left(\frac{-E_a}{RT}\right) t^z \quad (4)$$

$$C_{loss,cyc} = A(I) \exp\left(\frac{-E_a + \alpha|I|}{RT}\right) (Q_{Ah})^z \quad (5)$$

Although labeled similarly, all parameters for calendar and cycle aging result in different values, which have to be recalculated for every cell chemistry by doing extensive accelerated aging tests. Focusing more on hybrid electric vehicles (HEVs) other influences on the minimum state of charge SOC_{min} and operation mode *Ratio* were used in [7] as described in (6).

$$C_{loss} = (a + b \text{Ratio}^c + d(SOC_{min} - SOC_0)^e) \cdot \exp\left(\frac{-E_a}{RT}\right) (Q_{Ah})^z \quad (6)$$

Ratio is defined between 0 and 1 depending on the time proportion of discharge (pure electric driving) and charge sustaining in a plug-in HEV. A *Ratio* of 1 indicates a pure discharge with no charge sustaining. Many publications are available to improve specific effects like varying temperatures [35,36] or load profiles [37]. So, in general, all of the

presented ampere-counting empirical aging equations formulate similar dependencies, observing a square root time and/or charge throughput influence. Many papers account this to a side reaction taking place to build up the SEI. Furthermore, an Arrhenius-like temperature influence with the unknown parameter being the activation energy is used. Voltage or SOC is described to influence calendar aging and DOD to effect cycle aging. Although empirical equations are estimated based on observations and fitted to the measurement data, some similarities in the SPM aging equation can be found.

3. Single particle model with SEI layer growth

In contrast to an EEM, a SPM not only models current, voltage and temperature, but describes internal processes leading to concentration shift, transportation processes, and chemical reactions. Therefore, it provides a deeper insight and can describe more aging effects than an EEM. Still, if compared to multiphysics models, a SPM applies already many simplifications. For example, the SPM is only a one dimensional model and the electrodes themselves are assumed to consist of one particle, as displayed in Fig. 1 (a). However, a SPM is able to represent most of the dominant aging effects. The basic equations for the base, thermal and aging model are introduced in the following. For a more detailed description of the equations refer to [11,38,30,39]. An overview of the used variables is provided in Appendix C.

3.1. Base and thermal model

The equations of the average SPM are displayed in Fig. 1 (b). Voltage $V(t)$ is calculated using the demanded current $I(t)$, which is responsible for changes in the solid and liquid phase (electrolyte) concentration $c_{s,i}$ and c_e . Furthermore, a kinetic transport of Li-ions due to kinetic overpotential η_i^k and liquid phase potential ϕ_e at the electrodes at position 0 and L is introduced. The subscript i represents n or p , indicating the negative and positive electrodes processes. The concentrations on the other hand lead to a half cell open circuit voltage (OCV) U_i . In total, the cell voltage is described by (7).

$$V(t) = U_p\left(\frac{c_{s,p}}{c_{s,p,max}}\right) - U_n\left(\frac{c_{s,n}}{c_{s,n,max}}\right) + \eta_p^k - \eta_n^k + \phi_e(L) - \phi_e(0) \quad (7)$$

The SOC and capacity C of the cell is evaluated using the normalized bulk Li-ion concentration θ_n^b for the concentrations in a fully charged (100%) and discharged cell (0%).

$$SOC = 100 \left(\frac{\theta_n^b - \theta_{n,0\%}^b}{\theta_{n,100\%}^b - \theta_{n,0\%}^b} \right) \quad (8)$$

Further, the capacity Q_i of both cell electrodes can be calculated using x and y as stoichiometric coefficients for the negative and positive electrode, δ_n and δ_p as negative and positive electrode thickness, $\epsilon_{s,i}$ as active material volume fraction, electrode plate area A , and Faraday constant F .

$$Q_n = F \delta_n \epsilon_{s,n} A c_{s,n,max} |x_{100\%} - x_{0\%}| \quad (9)$$

$$Q_p = F \delta_p \epsilon_{s,p} A c_{s,p,max} |y_{100\%} - y_{0\%}| \quad (10)$$

A simple lumped parameter energy balance is used as thermal model. Temperature T change is induced by the difference between generated (12) and dissipated (13) energy with cell mass m and heat capacity C_p . Heat is generated by irreversible influences, for example resistive losses, as well as reversible influences. Heat is dissipated by convection to the ambient air T_{amb} with cell specific area A and convective coefficient h_{conf} . In contrast to most EEM, both surface and internal temperature are calculated and the equations are coupled to the SPM using an Arrhenius-like relation.

$$\frac{dT}{dt} = \frac{1}{mC_p} (\varphi_{gen} - \varphi_{dis}) \quad (11)$$

$$\varphi_{\text{gen}} = - \left((V - (U_p - U_n))I + T \frac{d(U_p - U_n)}{dT} I \right) \quad (12)$$

$$\varphi_{\text{dis}} = h_{\text{conv}} A_{\text{cell}} (T - T_{\text{amb}}) \quad (13)$$

3.2. Aging model

As discussed before, the growth of the SEI layer is modeled as dominant aging process. The layer is build up during the initial cycling of the cell and consists of solvent products mixed with Li-ions. During operation, the SEI layer partly blocks the solvents from reaching the negative electrode. As a consequence a reaction with Li-ions from the negative electrode is prevented and less side products form. Latter would add to the layer itself and lead to a resistance increase and capacity fade. The side reaction can be described by modeling the solvent concentration c_{solv} inside the SEI layer as described in (14), which itself depends on the layer thickness δ_{SEI} and the current and temperature dependent diffusion coefficient D_{solv} . This side reaction induces a current i_s (15). It is calculated based on a concentration dependent exchange current density i_s^0 and an exponential function, which depends on the internal temperature T and potential U . Additionally to the Li-ion concentration at the negative particle surface $c_{s,n}^s$, the Faraday constant F and the previous solvent concentration at the surface of the negative electrode c_{solv}^* are introduced. The aging constant k_f' (16) accounts for neglected aging effects.

$$\frac{\partial c_{\text{solv}}}{\partial t} = D_{\text{solv}}(I, T) \frac{\partial^2 c_{\text{solv}}}{\partial r^2} - \frac{d\delta_{\text{SEI}}}{dt} \frac{\partial c_{\text{solv}}}{\partial r} \quad (14)$$

$$i_s = -i_s^0 \exp \left(-\frac{E_{\text{ak}}}{R} \left(\frac{1}{T_{\text{int}}} - \frac{1}{T_{\text{ref}}} \right) - \frac{\beta F}{RT_{\text{int}}} \left(\phi_{s,n} - \frac{\delta_{\text{SEI}}}{\kappa_{\text{SEI}} S_n} - U_s \right) \right) \quad (15)$$

$$i_s^0 = 2Fk_f' c_{\text{solv}}^* (c_{s,n}^s)^2 \quad (16)$$

Note that the quadratic choice of $(c_{s,n}^s)^2$ depends on observations more than on physical motivation. Other dependencies might occur for a different cell chemistry. For example, LMO cells might show additional aging effects at low SOC due to dissolution of manganese. Therefore, the LMO cell's behavior might be better represented by a function of fourth order [39]. For simplicity, a quadratic term is used as described in [30]. The SEI layer thickness is evaluated, using its molar mass M_{SEI} and layer density ρ_{SEI} .

$$\frac{d\delta_{\text{SEI}}}{dt} = -\frac{M_{\text{SEI}}}{2F\rho_{\text{SEI}}} i_s \quad (17)$$

The remaining capacity C_{rem} (18) and resistance increase $R_{s,n} + R_{\text{SEI}}$ (19) can be calculated. The variable Q_s describes the capacity loss due to SEI growth, which depends on the negative electrode's thickness δ_n , surface area A , base resistance $R_{s,n}$ and active material volume fraction $\epsilon_{s,n}$. The resistance increase furthermore depends on the SEI ionic conductivity κ_{SEI} . In Fig. 1 (b) an overview of the aging equations and their impact on the electrical equations is highlighted in red.

$$\frac{d}{dt} Q_s = \frac{3\epsilon_{s,n}\delta_n A}{R_{s,n}} i_s$$

$$C_{\text{rem}}(t) = 100 \frac{Q_n - Q_s(t)}{Q_n} \quad (18)$$

$$\frac{dR_{\text{SEI}}}{dt} = -\frac{M_{\text{SEI}} R_{s,n}}{6F\rho_{\text{SEI}} \epsilon_{s,n} \delta_n A \kappa_{\text{SEI}}} i_s. \quad (19)$$

4. SPM derived EEM motivation

The aging equations of the EEM are developed based on knowledge of electrochemical models but mostly based on aging observations. To fill in the gap between electrochemical motivation and empirical

equations, it is shown – although by using some simplifications – that SPM aging equations lead to a similar expression as the empirical aging models.

Taking a look at the SPM model presented before, the equations can be combined and rearranged to derive a simplified relation. By inserting (15) in (18) and combining constant terms, the following representation is obtained:

$$\begin{aligned} \frac{d}{dt} C &= \frac{1}{Q_n} \frac{d}{dt} Q_s = \frac{1}{Q_n} \frac{3\epsilon_{s,n}\delta_n A}{R_{s,n}} i_s \\ &= -\zeta_4 c_{\text{solv}}^* (\theta_n^s)^2 \\ &\quad \cdot \exp \left(-\frac{1}{T_{\text{int}}} \left(\zeta_1 + \zeta_3 ((\phi_{s,n} - U_s) - \frac{\delta_{\text{SEI}}}{\kappa_{\text{SEI}} S_n}) \right) \right). \end{aligned} \quad (20)$$

In (20), the normalized inserted Li-ion concentration at the surface $\theta_n^s = c_{s,n}^s / c_{s,n,\text{max}}^s$ is introduced. Terms, which are material specific and therefore do not change, are simplified by introducing the constants ζ_j . For a detailed description of the derivation, refer to Appendix A.

4.1. Calendar aging

Assuming pure calendar aging, the equations can be further simplified as $I = 0$, and the concentrations will be in a steady state leading to $c_{s,n}^s = \text{constant}$. Furthermore, temperature changes are neglected, as the internal cell temperature will only change if the ambient temperature is changing (which is not the case in the aging tests conducted for empirical aging evaluation).

$$\frac{d}{dt} C_{\text{cal}} = -\zeta_4 c_{\text{solv}}^* (\theta_n^s)^2 \exp \left(-\frac{\zeta_1}{T_{\text{int}}} \right) \exp \left(-\frac{\zeta_3}{T_{\text{int}}} (\phi_{s,n} - U_s) \right) \quad (21)$$

Assuming that only the solvent concentration at the interface between SEI and negative electrode c_{solv}^* is changing during calendar aging, which is true if a constant voltage and temperature is maintained or the changes are negligible small, the total capacity loss due to calendar aging can be reformulated as

$$\begin{aligned} C_{\text{cal}}(t) &= C_0 - \zeta_4 (\theta_n^s)^2 \cdot \exp \left(-\frac{\zeta_3}{T_{\text{int}}} (\phi_{s,n} - U_s) \right) \\ &\quad \cdot \exp \left(-\frac{\zeta_1}{T_{\text{int}}} \right) \cdot \int_{\tau=0}^t c_{\text{solv}}^*(\tau) d\tau \end{aligned} \quad (22)$$

Determining the solvent concentration at the surface c_{solv}^* remains a main challenge. Furthermore, the partial differential equation (14) is hard to solve analytically. Therefore, SPM results are analyzed in order to establish the following relationships. As for example described in [21,40,29], the solvent concentration leads to a square root like SEI layer growth over time. Although they use some simplifications – [40] assumes a solvent diffusion equation neglecting the direct influence of the SEI layer growth (see (14)) – the fundamental assumption of a square root like layer increase holds. The formula can be written as:

$$c_{\text{solv}}^*(t) = \zeta_{\text{solv},1} \frac{1}{\sqrt{t + \zeta_{\text{solv},2}}} + \zeta_{\text{solv},3}. \quad (23)$$

Eq. (23) and (22) can be integrated and rewritten to look similar to the empirical relations given in (1) or the first term in (3):

$$\begin{aligned} C_{\text{loss,cal,SPM}} &= -\zeta_{\text{cl}} (\theta_n^s)^2 \cdot f(T, (\phi_{s,n} - U_s)) \\ &\quad \cdot \exp \left(-\frac{\zeta_1}{T_{\text{int}}} \right) \cdot \left(\sqrt{t + \zeta_{\text{solv},2}} + \zeta_{\text{cl}} t \right). \end{aligned} \quad (24)$$

A detailed derivation of this dependency is given in Appendix A.1. If it is assumed that $\text{SOC} \doteq \theta_n^s$, $V \doteq \phi_{s,n}$, and temperature influences on the voltage term are negligibly small, the following simplified SPM–EEM equation holds:

$$\begin{aligned} C_{\text{loss,cal,SPM-EEM}} &= -\zeta_{\text{cl}} f(\text{SOC}^2, V - U_s) \exp \left(-\frac{\zeta_1}{T_{\text{int}}} \right) \\ &\quad \cdot (\sqrt{t} + \zeta_{\text{cl}} t) \end{aligned} \quad (25)$$

Similarities can be observed by comparing the SPM–EEM equation (25)

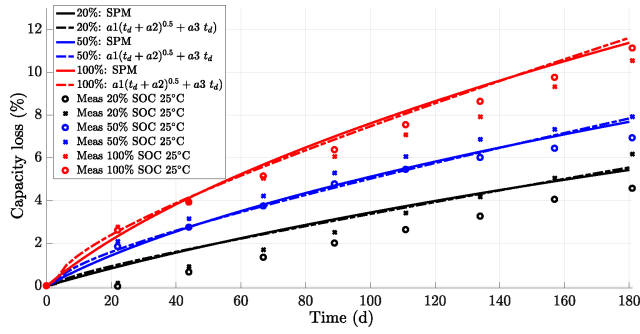


Fig. 2. Measured capacity loss (markers) versus capacity loss estimated by SPM (line) as well as time-dependent functions fitted to SPM data (dashed) for $T = 25^\circ\text{C}$ and $\text{SOC} = 20\%$ (black), 50% (blue), and 100% (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to the EEM calendar aging represented in [33,5], which is repeated for clarity:

$$C_{\text{loss,cal,EEM}} = a_1(V + a_2)\exp\left(-\frac{a_3}{T_{\text{int}}}\right)\sqrt{t}. \quad (26)$$

In (25) and (26) the Arrhenius law is used for temperature modeling. The SPM–EEM shows a more complex dependency on voltage than the linear term for the EEM. However, for small voltage changes, the exponential term can be approximated by its series expansion and a linear expression showing a $(V - U_s)$ -dependency is obtained. The derived simplified SPM–EEM equation (25) shows not only a square root-like development over time but additionally linear influences. This corresponds well with the SPM predicted aging and the measured capacity loss of the LMO cell (markers) depicted in Fig. 2. The original SPM results for 25°C and 20% , 50% , and 100% SOC are marked as solid lines and the fit based on the derived simplified equation (24) as dashed lines.

In literature, time dependency is often noted by a term t^z with $z = 0.5$. However, some authors such as [5] reported that $z = 0.75$ shows better accordance with the measurements. Linear and square-root like time dependency can be approximated by a term with z being approx. 0.75, leading to the observed behavior.

4.2. Cycle aging

In case of cycling, I is not 0 and can, therefore, not be neglected. Eq. (23) and (20) result in

$$\frac{d}{dt}C = -\zeta_4 c_{\text{solv,init}}^* (\theta_n^s)^2 f(T, \phi_{s,n} - U_s) \exp\left(-\frac{\zeta_1}{T_{\text{int}}}\right) \cdot \exp\left(\frac{\zeta_3}{T_{\text{int,avg}}} \frac{\delta_{\text{SEI}} I}{\kappa_{\text{SEI}} S_n}\right) \left(1 - \frac{\zeta_{\text{solv}}}{c_{\text{solv,init}}^* \sqrt{t}}\right) \quad (27)$$

and

$$C_{\text{loss,cyc}} = \zeta_4 c_{\text{solv,init}}^* \int (\theta_n^s)^2 f(T, \phi_{s,n} - U_s) \exp\left(-\frac{\zeta_1}{T_{\text{int}}}\right) \cdot \exp\left(\frac{\zeta_3}{T_{\text{int,avg}}} \frac{\delta_{\text{SEI}} I}{\kappa_{\text{SEI}} S_n}\right) \left(1 - \frac{\zeta_{\text{solv}}}{c_{\text{solv,init}}^* \sqrt{t}}\right) dt. \quad (28)$$

If constant cycling is assumed the equation can be simplified. Although voltage and temperature varies during cycling, these dependencies might then be represented by their averages. The normalized inserted Li-ion concentration can be regarded as a function of the DOD. In a constant charging and discharging scenario, the current applied to the cell is either I_0 or $-I_0$. Due to the monotonic increase of the exponential function, an upper bound is defined by taking only the positive current I_0 into account. Further, the series expansion of the exponential function is used. A detailed derivation of this result and the used

simplifications is provided in Appendix A.2. Although these are quite strong assumptions and simplifications and not all assertions have been proven, the results in Section 6 show that an empirical aging relation can be deduced in this way. In total, (28) can be reformulated and estimated as

$$C_{\text{loss,cyc,SPM}} \leq \zeta_{c1} f(\text{DOD} + \zeta_{e,n,2})^2 \cdot f(\phi V - U_s) \cdot \exp\left(-\frac{\zeta_1}{T_{\text{int}}}\right) \cdot (t - \zeta_{c2} \sqrt{t}) + \zeta_{c10} f(\text{DOD} + \zeta_{e,n,2})^2 \cdot f(\phi V - U_s) \cdot \exp\left(-\frac{\zeta_1}{T_{\text{int}}}\right) \cdot \frac{I_0 t (1 - \zeta_{c2} t^{-0.5})}{Q_{\text{Ah}}}. \quad (29)$$

The SPM aging equation (29) can be transformed into a simpler SPM–EEM aging equation (30) by neglecting the DOD influence on the quasi-calendar aging part and temperature influence on cycle aging.

$$C_{\text{loss,cyc,SPM-EEM}} = a_1(t + a_2 \sqrt{t}) \cdot (V + a_3) \cdot \left(\exp\left(\frac{a_4}{T}\right) + I_0 \cdot a_5 (\text{DOD} + a_6)^2\right) \quad (30)$$

In (30), a_j with $j = 1, 2, \dots, 6$ are constants, which have to be estimated or fitted according to measurements. SPM–EEM equation (30) can be used as an empirical equation for aging estimation. A comparison of this novel SPM–EEM aging relation to SEM and a conventional EEM aging model is shown in the following sections.

Still, the simplifications used to derive this empirical relation are only true for constant cycling situations as conducted during accelerated aging tests. As empirical aging models are also commonly fitted using such accelerated aging tests, (30) might be able to represent the aging behavior well. In case of pure calendar aging, the second expression of (29) vanishes and the expression as derived in (24) is obtained.

As a first step, this shows, that the empirical aging models based on ampere-counting can systematically be derived from SPM SEI layer growth. Although many researchers motivate and check the plausibility of their empirical models by referring to the square root like SEI layer growth, no approach to derive the empirical aging equations has been shown yet. Using the derived empirical aging model, an aging representation is added to an electrical equivalent model as described in the next section.

5. Electrical equivalent model and empirical aging

In most automotive applications, merely the current-voltage dependency of the battery is modeled using simple electrical equivalent circuits. The battery is modeled by a voltage source and at least one series resistance. If dynamic effects are taken into account, various additional resistors and capacitors, so-called RC-elements, or more advanced electric parts are added. Such a Randles circuit [41] is exemplarily shown for two RC-elements in Fig. 3. To account for aging effects, those EEM are commonly extended by empirical aging equations. As no internal cell parameters are known, those aging models are usually derived from observations obtained by accelerated aging tests and basic physical coherences. For example, see [5,7,34,10] for details. As a next step, an EEM with empirical aging equations following the scheme of [5,10,42] is introduced using the derived SPM–EEM aging relations (30).

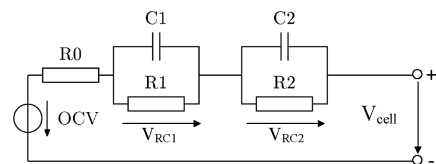


Fig. 3. Second order Randles circuit.

5.1. Base and thermal model

As electrical representation of the battery cell's behavior, a second order Randles circuit is chosen as seen in Fig. 3. The resulting voltage $V(t)$ is derived using Kirchhoff rules with the open circuit voltage V_{OCV} , current $I(t)$, voltage over the RC-elements V_{RC} as well as series resistances R_0 , R_1 , R_2 and capacitors C_1 and C_2 .

$$V(t) = V_{OCV} + R_0 I(t) + V_{RC1}(t) + V_{RC2}(t) \quad (31)$$

$$\frac{dV_{RC1,2}(t)}{dt} = \frac{-1}{R_{1,2}C_{1,2}} V_{RC1,2}(t) + \frac{1}{C_{1,2}} I(t) \quad (32)$$

For simulation purposes, the SOC can be determined using ampere counting starting from an initial SOC_{init} .

$$SOC(t) = SOC_{init} + \int_{t_0}^t I(\tau) d\tau \quad (33)$$

As thermal model, a simple lumped parameter approach is used, where heat is generated by resistive losses and dissipated by convection. Temperature T changes are induced based on cell mass m , heat capacity c , convective constant h_{conv} , cell area A , and ambient temperature T_{amb} .

$$\frac{dT}{dt} = \frac{1}{mc} (IV_{OCV} - VI - h_{conv}A(T - T_{amb})) \quad (34)$$

5.2. Aging model

As described before, simple aging models are based on empirical equations derived from observations and accelerated aging measurements. They depend of the EEM parameters such as time, voltage, temperature, current and cycle depth. Different models and dependencies are described in literature. In this case the derived aging equation (30) is used. Further, an existing empirical aging equation developed by [5] is shortly introduced and used to assess the quality of the model.

In the model presented by [5,42], aging is divided into a calendar and cycle aging part. During storage, aging shows a square-root-like time t dependency, which is influenced by the voltage level and temperature during storage. Cycle aging depends on the charge throughput Q , average voltage $\bar{V}$ and depth of discharge DOD . Both, capacity loss C_{loss} and resistance increase R_{inc} are described in [5]. In this study, only the aging relations describing the capacity loss are evaluated.

$$C_{loss} = \overbrace{\alpha_c(V, T) t^{0.75}}^{\text{calendar}} - \overbrace{\beta_c(\bar{V}, DOD) \sqrt{Q}}^{\text{cycle}} \quad (35)$$

$$\alpha_c(V, T) = a_1(V + a_2) \exp\left(\frac{a_3}{T}\right) \quad (36)$$

$$\beta_c(\bar{V}, DOD) = b_1(\bar{V} + b_2)^2 + b_3 DOD + b_4. \quad (37)$$

In case of capacity increase, a linear behavior on voltage is reported. Aging increases linearly with increasing voltage. Temperature dependency can be estimated using an Arrhenius-like representation. When cycling, low as well as very high average voltages should be avoided, leading to a parabolic dependency. Higher DOD s additionally lead to faster aging. The parameters a_i and b_i with $i = 1, 2, 3, 4$ have to be identified using aging measurements and have no direct physical meaning.

Comparing this to the derived aging relation (30), which is stated again for clarity, similarities and differences can be observed.

$$C_{loss, SPM-EEM} = a_1(t + a_2\sqrt{t}) \cdot (V + a_3) \cdot \left(\exp\left(\frac{a_4}{T}\right) + I_0 \cdot a_5(DOD + a_6)^2 \right) \quad (38)$$

In the calendar aging case, I_0 becomes 0 and

$$C_{loss, SPM-EEM} = a_1(V + a_3) \cdot \exp\left(\frac{a_4}{T}\right) (t + a_2\sqrt{t}) \quad (39)$$

Hence, calendar aging depends on the same influencing factors as in (36). The only difference is the time dependency. However, as shown before, $t^{0.75}$ and $t + a_2\sqrt{t}$ lead to a similar relation. When neglecting the calendar aging part to see cycling specific effects, on the other hand, (38) becomes

$$C_{loss, SPM-EEM} = a_1(V + a_3) \cdot a_5(DOD + a_6)^2 \cdot I_0 \cdot (t + a_2\sqrt{t}) \quad (40)$$

Compared to (37), voltage dependency is not quadratic but DOD . And instead of a sum, both factors multiply in the SPM-EEM equation. The term $I_0(t + a_2\sqrt{t})$ corresponds to the charge throughput Q_{Ah} and shows also both linear and square root dependencies on that factor. In total, the SPM-EEM equations correspond to already derived EEM empirical aging relations, which were mostly proven by fitting the parameters to measurements. This indicates, that such empirical relations do indeed represent the chemical processes occurring, although in a simplified manner.

In order to motivate the benefits and drawbacks of such simple aging models compared to an electrochemical representation using the SPM, both described models are compared for adaptability, validity, and performance in the next section.

6. Adaption to LMO cell chemistry

After the basic model equations are known, they have to be adapted as every Li-ion cell chemistry shows different charge and discharge characteristics as well as aging behavior. For comparison purposes a test campaign covering calendar and cycle aging tests was conducted. Prismatic LMO cells with a nominal capacity of 12 Ah were purchased. Capacity and resistance of the cells were determined based on a 1 C discharge. The supplier defined voltage range is between 2.7 V and 4.2 V. All tests were conducted at 1 C charge and discharge rate. LMO cells were stored at 20%, 50%, and 100% state of charge (SOC) at 25 and 45 °C. Cycling was done at 25 °C at 5–100%, 25–45%, and 25–75% SOC. After 3 weeks of storage and cycling, remaining capacity and resistance was determined at 25 °C and 1 C discharge rate. A description of those tests and results can also be found in [39,42]. Results of this test campaign are used to fit both an EEM with empirical aging equation and a SPM. For calendar aging, results are presented over time and for cycle aging equivalent full cycles are used to compare results of different cycle depth.

One of the main advantages of EEM with empirical aging equations is the simpler parametrization in case voltage, temperature, and aging characteristics are available. Less fit parameters are to be determined and no cell internal parameters are required. The steps to parametrize a new cell for use with the SPM-EEM and SPM are shortly shown in the following.

6.1. SPM-EEM parametrization

For EEMs, fit values for the components of the electric circuit, thermal model, as well as fits for the aging representation have to be identified.

6.1.1. Electric and thermal fit

The electric parameters can be identified using one charge and discharge cycle or impulses. Similarly thermal model parameters can be fit. Exemplarily, this results in resistances $R_0 = 7 \text{ m}\Omega$, $R_1 = 0.6 \text{ m}\Omega$, and $R_2 = 6 \text{ m}\Omega$, as well as capacities of $C_1 = 4600 \text{ F}$, and $C_2 = 9100 \text{ F}$. The heat model parameters are the effective surface area $A = 0.035 \text{ m}^2$ and heat capacity $c = 900 \text{ J/(kg K)}^{-1}$, heat transfer coefficient $h_{conv} = 10 \text{ W m}^{-2} \text{ K}^{-1}$ and cell mass $m = 0.4 \text{ kg}$. The results are displayed in Fig. 4. Those static measurements can be extended to characteristic maps covering temperature, SOC and resistance depending effects.

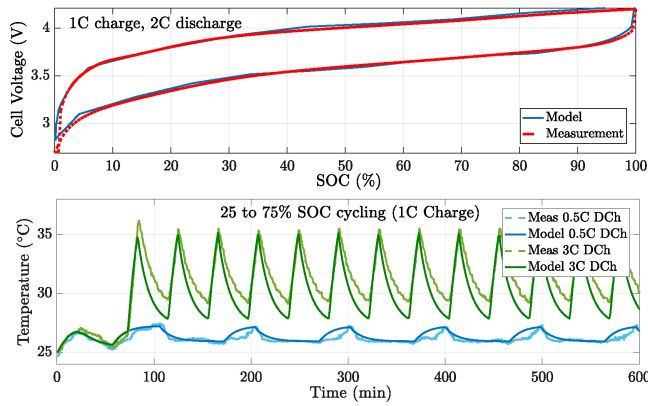


Fig. 4. EEM base and thermal model fit.

6.1.2. Aging fit

As aging model both the novel SPM–EEM aging relation (38) and the model presented by [5] are implemented on top of the base and thermal model.

For identification of the SPM–EEM model parameters, the model was fit to the calendar aging curves at first. This resulted in parameter estimates on the influence of the average voltage a_3 and activation energy a_4 for the Arrhenius term. Then, those values were used to fit the model to the cycle aging scenarios and to identify a_1 , a_2 , a_5 and a_6 . All estimations were done in a least-squares sense.

$$C_{\text{loss}} = 4.34 \cdot 10^{-7} (t + 35\sqrt{t}) \cdot (V - 3.23) \cdot \left(\exp\left(\frac{-5010}{T}\right) + I_0 \cdot 0.64 (\text{DOD} + 63.7)^2 \right) \quad (41)$$

For the aging model provided by [5], calendar constants $a_{1,2,3}$ and cycle constants $b_{1,2,3,4}$ need to be calculated. Using (35) together with (36) and (37), the values can be fit to the accelerated aging measurements. This can either be done simultaneously for all cell parameters and measurements, or at first for calendar and then cycle aging, as described in [5].

6.2. SPM parametrization

The SPM model consists of more complex equations as the internal processes are also build up. Therefore, it requires more parameters which are harder to identify. In contrast to EEM, the parameters cannot be directly fit to measurement data, but need to be measured or obtained from literature data. Since no test equipment, time for cell disassembling, and half cell measurements are usually available, the parameters are taken from literature data. Although the cell chemistry is known, different values for LMO cells are reported, which are a result of different electrode compositions, separator and electrolyte materials. Using some exemplary values, those difficulties are shortly explained in the following.

6.2.1. Base and thermal fit

As most Li-ion cells use graphite as negative electrode material, the basic values describing particle size and thickness are easy to obtain. Still, parameters, which depend also on the positive electrode material, show greater variance. In Table 1, some of the basic parameters are presented. As data source, values obtained from LMO cells by [32,43–45] and a LiFePO cell by [38,30] are used. Values from other cells than LMO are used as not all parameters were available in [32,43–45]. The origin of these values, namely measured, estimated, adjusted, or from cell design procedure, is displayed. In case of the active material volume or porosity, differences in the positive electrode of LMO and LFP cells can be clearly recognized. Especially for the negative electrode, most reported values match closely. Parameter values representing chemical and physical interactions, such as the exchange

Table 1

Comparison of literature values for LFP and LMO cells.

Parameter	Source	Unit	pos. electr.	neg. electr.
Electrode thickness δ			$6.2 \cdot 10^{-5}$ (e)	
	[38]	m	$8.0 \cdot 10^{-5}$ (m)	$8.4 \cdot 10^{-5}$ (m)
	[32]		$3.6 \cdot 10^{-5}$	$5.0 \cdot 10^{-5}$
	[43]		$8.0 \cdot 10^{-5}$	$4.5 \cdot 10^{-5}$
	[44]		$3.6 \cdot 10^{-5}$ (d)	$4.3 \cdot 10^{-5}$ (d)
Active material volume R_s			0.538 (e)	0.624 (e)
	[38]	–	0.374 (a)	0.580
	[32]		0.500	0.580
	[44]		0.559 (d)	0.566 (d)
	[30]		0.426 (e)	0.360
Porosity ϵ_e	[32]	–	0.330	0.332
	[43]		0.300	0.330
	[44]		0.330 (d)	0.380 (d)
Max. solid phase conc. $c_{s,max}$	[38]		22,806	30,555
	[44]	mol m^{-3}	23,230	27,362
	[45]		22,860	30,540
Exchange current density i_0			1.6 (e)	1.6 (e)
	[30]	A m^{-2}	0.05 (e)	7.5 (e)
	[45]		0.416	0.688
Solid phase Li diffusion D_s	[38]		$5.9 \cdot 10^{-17}$ (a)	$8.0 \cdot 10^{-12}$
	[43]	$\text{m}^2 \text{s}^{-1}$	$2.5 \cdot 10^{-15}$	$9.0 \cdot 10^{-14}$
	[44]		$4.0 \cdot 10^{-14}$ (e)	$1.1 \cdot 10^{-14}$ (e)

Listed values are (a)adjusted, (e)estimated, (m)asured, parameters obtained by cell (d)esign or no note if data source is unknown.

Table 2

Thermal model parameter values.

Parameter	Symbol	Unit	Value
Convective coefficient	h_{conv}	$\text{W K}^{-1} \text{m}^{-2}$	10
Heat capacity	C_p	J (kg K)^{-1}	900
Cell lateral surface	A_{cell}	m^2	0.0035
Cell mass	m	kg	0.4

current density, show great variance. This complicates a precise model formulation. Thermal model parameters also depend on the cell size and chemistry, but can be obtained easier. Exemplary values are displayed in Table 2. Once the cell specific parameters are chosen and fit, the SPM can resemble the electrical and thermal behavior as seen in Fig. 5.

6.2.2. Aging fit

Finding an adequate data base for parametrization of the aging model becomes even harder. Not much literature deals with aging behavior of electrochemical models and even fewer present their used

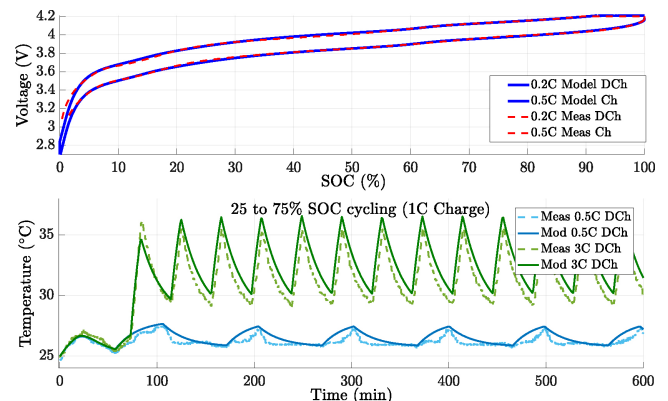


Fig. 5. SPM base and thermal model fit.

Table 3
Overview of aging model parameters.

Parameter	Source	Unit	Neg. electr. value
Initial SEI thickness $\delta_{SEI}(0)$	[30] [44]	m	$5.0e^{-9}$ (e) $4.5e^{-7}$
Solvent reduction kinetic const. k_f'			$0.85e^{-22}$ (cal)
		$m^7 (s \text{ mol})^{-1}$	$46.75e^{-22}$ (cyc)
	[30]		$1.18e^{-22}$ (cal)
	[30]		$141.6e^{-22}$ (cyc)
SEI layer molar mass M_{SEI}	[30]	$kg \text{ mol}^{-1}$	0.162
SEI layer density ρ_{SEI}	[30]	$kg \text{ m}^{-3}$	1690
SEI ionic conductivity κ_{SEI}	[30] [44]	$S \text{ m}^{-1}$	$17.5e^{-5}$ (a) $24e^{-5}$
SEI porosity ϵ_{SEI}		%	1.5 (cal)
	[30]		1 (cal)
Initial solvent diffusion coeff. D_{sol}^0			$8.58e^{-19}$ (cal)
		$m^2 s^{-1}$	$8.84e^{-20}$ (cal)
	[30] [44]		$2.85e^{-16}$ (cal)

Listed values are (a)adjusted, (e)stimated, or no note if data source is unknown. In case parameters are obtained from (cal)endar and (cyc)le aging tests, those are marked.

measurements and assumptions. Furthermore, some of the parameters, such as the solvent reduction kinetic constant k_f' and SEI porosity ϵ_{SEI} , need to be adapted based on calendar and cycle aging measurements. Table 3 presents a short overview of data sets available. The adaption process is also discussed in [39] in more detail.

If the parameters are derived, the resulting aging behavior

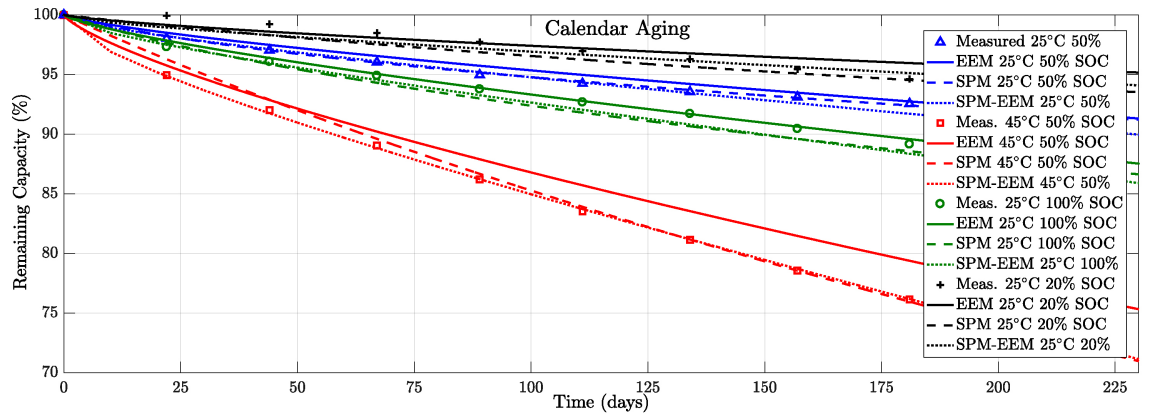
resembles the obtained real aging data quite well as displayed in Fig. 6. Both calendar and cycle aging capacity decrease show a good match. The SPM capacity was also only calculated at a step size of about 200 equivalent full cycles as the model has to be stopped in between to calculate the cell capacity during cycle aging.

6.3. Model comparison

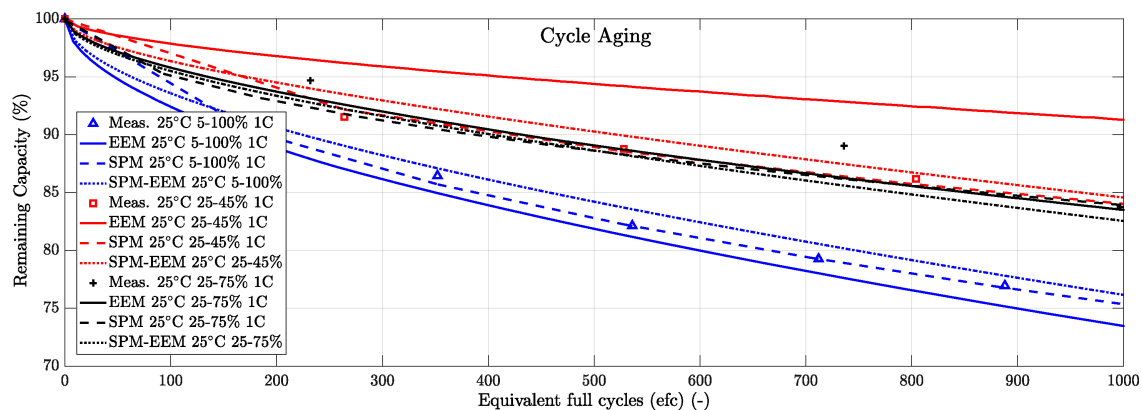
As described before, parametrization of an EEM is easier than for a SPM. An EEM usually consists of less than 10 parameters for the base and thermal model, depending on the used electrical equivalent representation. There is no uniform empirical aging model at the moment, but the described SPM–EEM aging model shows good accordance with some cells presented in literature such as [34,5].

A SPM on the other hand needs intensive parametrization due to internal cell parameters and transportation processes. Although most values are determined by physical and chemical relations, they are either hard to measure or cannot be measured at all. Every cell chemistry, type of electrolyte and separator influences the internal processes and thus need known or identified. In total, this leads to more than 60 parameters for the base and thermal model. Additionally, more than 15 parameters are required for the SEI aging extension, some of which are dependent on current and SOC.

The resulting aging behavior in comparison to the measurements obtained by an accelerated aging test campaign is shown in Fig. 6. At the top, pure calendar aging for a SOC of 20, 50 and 100% is displayed. The SPM–EEM aging model (dotted line) and EEM model presented by [5] (solid line) show good accordance with the measured capacity loss (markers). The first is able to represent higher temperatures better and



(a)



(b)

Fig. 6. EEM, SPM, and SPM–EEM aging model results in comparison to measurements: (a) calendar aging and (b) cycle aging.

the second has a better fit at 100% SOC and 25 °C. In average, the sum of the squared deviations for both empirical aging models and all measured SOC and temperatures shows only small differences. Results obtained by the SPM are displayed as dashed lines. The SPM has a better squared deviation than the SPM–EEM and EEM. However, at the same time the computation time and effort for the SPM is much higher. In total, all methods provide good results for calendar aging and vary only at computation time and parametrization effort.

Fig. 6 (b) provides results for cycling between 25 and 45% SOC, 5 to 100%, and 25 to 75%. The advantage of the SPM can be observed in this figure. As more complex processes occur during cycling, empirical capacity estimates cannot represent all of those. Whereas EEM and SPM–EEM are able to predict the full cycling (5–100%) with an acceptable error, the EEM aging relation used by [5] (solid line) has difficulties predicting the 25–45% cycling, as a linear influence on DOD is assumed. Therefore, the cell cycled between 25 and 45% cannot be properly modeled. This cell shows similar aging as the cell cycled between 25 and 75%, but the equations would indicate that the aging has to be lower due to the smaller DOD and lower average SOC. The SPM–EEM (dotted line) is able to predict this scenario better but is bound by the assumptions stated in the empirical equations. Therefore, larger deviations occur at other cycle depths.

In general, EEM, SPM–EEM, and SPM are able to represent the voltage and thermal behavior of the cell. The root mean square error RMSE and normalized RMSE are used to quantify the aging model fit. Therefore, the deviation of measurement y_n of every aging scenario n is compared to the model prediction $\hat{y}_n$ over all time instances T (for calendar aging) and equivalent full cycles (for cycling).

$$\text{RMSE} = \sqrt{\frac{\sum_{t=1}^T (\hat{y}_n(t) - y_n(t))^2}{T}} \quad (42)$$

The average measurement value $\bar{y}$ is used for presenting a normalized RMSE.

$$\text{RMSE}_{\text{norm}} = \frac{\text{RMSE}}{\bar{y}} \quad (43)$$

Table 4 presents the RMSE results for calendar and cycle aging and the overall model fit. For calendar aging all models and the measurements show good accordance with the SPM and especially the SPM–EEM being more accurate. The advantage of the SPM can be observed during cycle aging. Whereas the EEM and SPM–EEM are restricted to the modeled linear, exponential and quadratic aging dependencies, the SPM is also able to resemble more complex behaviors. Still, with an RMSE of 1.44, the SPM–EEM performs much better than the EEM (3.66). In total, the SPM–EEM performs only 15% worse than the SPM.

Besides the amount of variables, parametrization effort, and accuracy, the calculation time plays an important role when choosing a model. In the presented case, a comparison between the models, which were implemented in Matlab, was done using a standard working laptop. For a simulation time of 180 days, pure calendar and continuous fixed discharge cycle aging calculations were executed. For pure calendar aging, the SPM–EEM and EEM finish in less than 1 min and the SPM in about 10 min. For continuous cycling the SPM–EEM and EEM

simulation requires at most 25 min (using derivatives of the aging relation to be able to adapt for changing temperature and cycle depth), whereas the SPM runs for at least 7 h covering the same scenario. Although the SPM takes about 15 times longer than the EEM, this is, for simulation purposes, acceptable. On the other hand, in combination with a vehicle model and an optimization strategy this might be costly. Further, a SPM requires PDEs to be solved whereas the SPM–EEM and EEM uses algebraic equations and ODEs. This provides challenges for SPM implementation on control devices.

In total, the described SPM provides an insight into cell concentrations and degradation behavior. In comparison to other electrochemical models, the complexity of SPMs is acceptable. Still, the SPM–EEM and conventional EEM show already a good electrical, thermal, and aging estimate, which is sufficient for most automotive applications. By deriving the empirical aging equations from the SPM, a novel SPM–EEM aging model is described. It provides slightly better results than the EEM aging relations described in literature. Additionally, the empirical relations might be further improved by neglecting less parameters. This would lead to more fit parameters and a slightly higher parametrization effort. As the empirical relations are algebraic expressions, adding more parameters will increase the computational load only slightly. Still, the EEM aging equations rely on basic electrical and thermal parameters. Therefore, this approach is limited since the electrical and thermal model cannot provide cell internal information such as concentrations and layer growth which the SPM uses.

7. Conclusions

Based on an EEM with empirical aging equations and an average SPM with SEI motivated aging, advantages and difficulties of an electrochemical cell aging model have been investigated. Starting with SEI layer growth modeling for SPM aging equations, an EEM aging representation can – although by applying some simplifications – be derived. Using this knowledge, the SPM–EEM aging equation showed similarities to already existing empirical relations but also other relationships for voltage and DOD. The SPM–EEM was able to improve the accuracy of the aging model. Further research can be done to add more SPM related aging effects to the SPM–EEM.

Whereas an EEM is easy to fit and adapt to another cell chemistry, a SPM requires more parameters, which are difficult to obtain. In the end, it depends on the application of the model whether an EEM with easy parametrization and fast calculation or a SPM with more insight into cell processes and a better accuracy is needed. For most automotive applications EEMs are the preferred choice because they are less complex and can be implemented on control devices. Therefore, improving the EEM aging prediction is an important task for online lifetime estimation as complexity and calculation time is not increased. However, for more and more offline scenarios such as battery lifetime optimization, SPMs are used to increase the predictability and to cover more effects. In this case, computers can be used and computation time and complexity is less an issue. As a next step, improved empirical aging relations can be derived based on the results. In case less simplifications are used, the SPM–EEM aging equation can be expanded to increase the predictability of EEM aging models.

Acknowledgements

This work was developed as a collaboration of the University of Stuttgart, Bosch (China) Investment Ltd., and the Tongji University Shanghai.

Table 4
Model error.

Model	EEM	SPM	SPM–EEM
RMSE cal	0.77	0.54	0.38
RMSE cyc	3.66	1.20	1.44
RMSE overall	1.93	0.67	0.77
RMSE norm	0.021	0.007	0.008

Appendix A. SPM motivated EEM transformation

The SPM model derived can be simplified by inserting (15) in (18) and combining constant terms. The following representation is obtained:

$$\frac{d}{dt}C = \frac{1}{Q_n} \frac{d}{dt}Q_s = \frac{1}{Q_n} \underbrace{\frac{3\epsilon_{s,n}\delta_n A}{R_{s,n}}}_{=S_n} i_s \quad (\text{A.1})$$

$$\begin{aligned} \frac{d}{dt}C = & -\frac{S_n}{Q_n} i_s^0 \underbrace{\exp\left(\frac{E_{ak}}{R} \frac{1}{T_{ref}}\right)}_{=\zeta_2} \cdot \exp\left(-\frac{E_{ak}}{R} \frac{1}{T_{int}}\right) \\ & \cdot \exp\left(-\frac{\beta F}{R} \frac{1}{T_{int}} (\phi_{s,n} - U_s)\right) \cdot \exp\left(-\frac{\beta F}{R} \frac{\delta_{SEI}}{T_{int} \kappa_{SEI} S_n} \frac{I}{S_n}\right) \end{aligned} \quad (\text{A.2})$$

$$\begin{aligned} \frac{d}{dt}C = & -\underbrace{\frac{S_n \zeta_2}{Q_n} 2Fk'_f(c_{s,n,max}^s)^2}_{=\zeta_4} c_{solv}^* (\theta_n^s)^2 \cdot \exp\left(-\frac{\zeta_1}{T_{int}}\right) \\ & \cdot \exp\left(-\frac{\zeta_3}{T_{int}} (\phi_{s,n} - U_s)\right) \cdot \exp\left(-\frac{\zeta_3}{T_{int}} \frac{\delta_{SEI}}{\kappa_{SEI} S_n} \frac{I}{S_n}\right) \\ = & -\zeta_4 c_{solv}^* (\theta_n^s)^2 \\ & \cdot \exp\left(-\frac{1}{T_{int}} (\zeta_1 + \zeta_3 ((\phi_{s,n} - U_s) - \frac{\delta_{SEI}}{\kappa_{SEI} S_n} \frac{I}{S_n}))\right). \end{aligned} \quad (\text{A.3})$$

In (A.3), the normalized inserted Li ion concentration at the surface $\theta_n^s = c_{s,n}^s / c_{s,n,max}^s$ is introduced. Terms, which are material specific and therefore do not change, are simplified by introducing the constants ζ_j with $j = 1$ to 7.

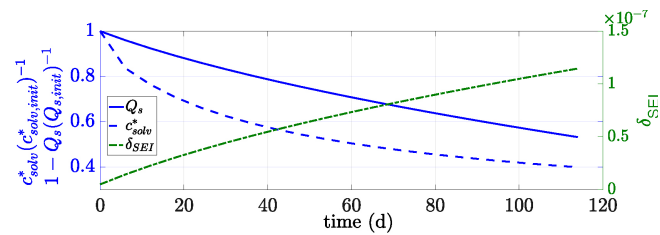
A.1 Calendar aging

Determining the solvent concentration at the surface c_{solv}^* remains a main challenge. The partial differential equation (14) is hard to solve analytically. Therefore, SPM results are analyzed in order to establish relationships. As for example described in [21,40,29], the solvent concentration leads to a square root like SEI layer growth over time. Although they use some simplifications – [40] assumes a solvent diffusion equation neglecting the direct influence of the SEI layer growth (see (14)) – the fundamental assumption of a square root like layer increase holds also in this context. This results from the solvent concentration at the interface between SEI and negative electrode c_{solv}^* . In Fig. A.7(a), c_{solv}^* is compared to SEI layer thickness δ_{SEI} and remaining capacity Q_s . The curves are taken from a SPM fitted to the described LMO cell as described in Section 6.

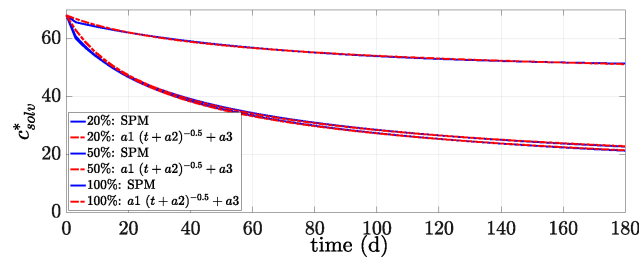
Changes of δ_{SEI} and Q_s depend on the current of the side reaction i_s as seen in (17) and (18), and, therefore, on c_{solv}^* . As displayed in Fig. A.7 (b), c_{solv}^* itself shows an inverse square-root-like behavior, which can be simplified as

$$c_{solv}^*(t) = \zeta_{c_{solv},1} \frac{1}{\sqrt{t + \zeta_{c_{solv},2}}} + \zeta_{c_{solv},3} \quad (\text{A.4})$$

and the integral becomes



(a)



(b)

Fig. A.7. (a) Square root like progress of solvent concentration at the interface between SEI and negative electrode c_{solv}^* normalized to initial concentration at start of calendar aging, SEI layer thickness δ_{SEI} and remaining capacity Q_s normalized to initial capacity taken from the SPM fitted to the LMO cell. Part (b) shows fits of a inverse square root curve for c_{solv}^* .

$$\int_{\tau=0}^t c_{\text{solv}}^*(\tau) d\tau = 2\zeta_{\text{cso},1} \sqrt{t + \zeta_{\text{cso},2}} + \zeta_{\text{cso},3} t \quad (\text{A.5})$$

Assuming capacity loss instead of remaining capacity for simplicity ($C_0 = 0$), (22) becomes

$$C_{\text{loss,cal}} = \zeta_4 \exp\left(-\frac{\zeta_1}{T_{\text{int}}}\right) (\theta_n^s)^2 \exp\left(-\frac{\zeta_3}{T_{\text{int}}} (\phi_{s,n} - U_s)\right) \cdot (2\zeta_{\text{cso},1} \sqrt{t + \zeta_{\text{cso},2}} + \zeta_{\text{cso},3} t). \quad (\text{A.6})$$

By substituting the constant terms with $\zeta_{c1} = \zeta_4 2\zeta_{\text{cso},1}$ and $\zeta_{c2} = \frac{\zeta_{\text{cso},3}}{2\zeta_{\text{cso},1}}$ and using the series expansion of the exponential function, the capacity loss over time is approximately

$$C_{\text{loss,cal}} = -\zeta_{c1} (\theta_n^s)^2 \cdot \exp\left(-\frac{\zeta_1}{T_{\text{int}}}\right) \left(\sqrt{t + \zeta_{\text{cso},2}} + \zeta_{c2} t \right) \cdot \left(1 + \left(-\frac{\zeta_3}{T_{\text{int}}} (\phi_{s,n} - U_s)\right) + \frac{1}{2} \left(-\frac{\zeta_3}{T_{\text{int}}} (\phi_{s,n} - U_s)\right)^2 + \dots \right). \quad (\text{A.7})$$

The dependency of the normalized Li concentration at the surface of the electrode on the SOC can be shown by including the boundary conditions:

$$\begin{aligned} \theta_n^s &= \frac{c_{s,n}^s}{c_{s,n,\text{max}}^s} = \frac{1}{c_{s,n,\text{max}}^s} \int \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 D_{s,n} \frac{\partial}{\partial r} c_s \right) \Big|_{r=R_{s,n}} dt \\ &= \frac{1}{c_{s,n,\text{max}}^s} \int \frac{1}{R_{s,n}^2} 2 R_{s,n} \left(\frac{-I}{F A_n} \right) dt \\ &= \frac{-2}{R_{s,n} F A_n c_{s,n,\text{max}}^s} \int I dt = \frac{-2}{R_{s,n} F A_n c_{s,n,\text{max}}^s} Q_c. \end{aligned} \quad (\text{A.8})$$

In (A.8), the charge throughput Q_c resembles the SOC of the cell. Furthermore, $\phi_{s,n}$ represents the open circuit voltage of the negative electrode in steady state. Both θ_n^s and $\phi_{s,n}$ attain values between 0 and 1 depending on the SOC and equivalently voltage V .

A.2 Cycle aging

Although voltage and temperature varies during cycling, some simplifications can be used in case constant cycling is assumed. In this case voltage and temperature dependencies are represented by their average ($\theta_n^s(t) = \varnothing \theta_n^s$, $\phi_{s,n}(t) = \varnothing \phi_{s,n}$, $T_{\text{int}}(t) = \varnothing T_{\text{int}}$), which is reasonable if the same cycle is repeated as in the aging tests. Further, the normalized inserted Li ion concentration at the surface depends linearly on the charge throughput as derived in (A.8). Therefore, its average can be described by $\varnothing \theta_n^s = 0.5(\theta_{n,\text{max}}^s - \theta_{n,\text{min}}^s) = \zeta_{\varnothing,n,1} \text{DOD} + \zeta_{\varnothing,n,2}$, with DOD being the depth of discharge. Eq. (20) can be rewritten as:

$$\begin{aligned} C_{\text{loss,cyc}} &= \zeta_4 c_{\text{solv,init}}^* (\zeta_{\varnothing,n,1} \text{DOD} + \zeta_{\varnothing,n,2})^2 \\ &\cdot f(\varnothing T_{\text{int}}, \varnothing \phi_{s,n} - U_s) \cdot \exp\left(-\frac{\zeta_1}{\varnothing T_{\text{int}}}\right) \\ &\cdot \int \exp\left(\frac{\zeta_3}{\varnothing T_{\text{int}}} \frac{\delta_{\text{SEI}} I(t)}{\kappa_{\text{SEI}} S_n}\right) \left(1 - \frac{\zeta_{\text{cso}}}{c_{\text{solv,init}}^* \sqrt{t}}\right) dt \end{aligned} \quad (\text{A.9})$$

During constant current discharging and charging with amplitude I_0 , the current $I(t)$ switches between I_0 and $-I_0$. As the exponential term is strictly monotonically increasing, an upper bound for capacity loss is provided when using $I(t) = I_0$. In this case the capacity loss is

$$\begin{aligned} C_{\text{loss,cyc}} &\leq \zeta_{c1} \zeta_{\varnothing,n}^2 \cdot \text{DOD}^2 \cdot f(\varnothing T_{\text{int}}, \varnothing \phi_{s,n} - U_s) \\ &\cdot \exp\left(-\frac{\zeta_1}{\varnothing T_{\text{int}}}\right) \cdot \exp\left(\frac{\zeta_3}{\varnothing T_{\text{int}}} \frac{\delta_{\text{SEI}} I_0}{\kappa_{\text{SEI}} S_n}\right) \\ &\cdot (t - \zeta_{c2} \sqrt{t}) \end{aligned} \quad (\text{A.10})$$

This term can be separated in a quasi-calendar aging and cycle aging part by using the series expansion of the exponential function:

$$\begin{aligned} C_{\text{loss,cyc}} &\leq \zeta_{c1} (\zeta_{\varnothing,n,1} \text{DOD} + \zeta_{\varnothing,n,2})^2 \cdot f(\varnothing T_{\text{int}}, \varnothing \phi_{s,n} - U_s) \\ &\cdot \exp\left(-\frac{\zeta_1}{\varnothing T_{\text{int}}}\right) \cdot (t - \zeta_{c2} \sqrt{t}) \\ &+ \zeta_{c1} \zeta_{\varnothing,n}^2 \cdot \text{DOD}^2 \cdot f(\varnothing T_{\text{int}}, \varnothing \phi_{s,n} - U_s) \\ &\cdot \exp\left(-\frac{\zeta_1}{\varnothing T_{\text{int}}}\right) \cdot (t - \zeta_{c2} \sqrt{t}) \\ &\left(\left(\frac{\zeta_3}{\varnothing T_{\text{int}}} \frac{\delta_{\text{SEI}} I_0}{\kappa_{\text{SEI}} S_n} \right) + \frac{1}{2} \left(\frac{\zeta_3}{\varnothing T_{\text{int}}} \frac{\delta_{\text{SEI}} I_0}{\kappa_{\text{SEI}} S_n} \right)^2 + \dots \right) \end{aligned} \quad (\text{A.11})$$

When neglecting higher order terms of the series expansion, inferences from SPM aging about empirical aging can be drawn.

$$\begin{aligned}
C_{\text{loss,cyc}} \leq & \zeta_{c1} (\zeta_{\theta,n,1} \text{DOD} + \zeta_{\theta,n,2})^2 \cdot f(\varnothing T_{\text{int}}, \varnothing \phi_{s,n} - U_s) \\
& \cdot \exp\left(-\frac{\zeta_1}{\varnothing T_{\text{int}}}\right) \cdot (t - \zeta_{c2} \sqrt{t}) \\
& + \zeta_{c1} \cdot (\zeta_{\theta,n,1} \text{DOD} + \zeta_{\theta,n,2})^2 \cdot f(\varnothing T_{\text{int}}, \varnothing \phi_{s,n} - U_s) \\
& \cdot \exp\left(-\frac{\zeta_1}{\varnothing T_{\text{int}}}\right) \cdot \left(\frac{\zeta_3}{\varnothing T_{\text{int}}} \frac{\delta_{\text{SEI}}}{\kappa_{\text{SEI}} S_n}\right) \\
& \cdot \frac{I_0 t}{Q_{\text{Ah}}} \left(1 - \zeta_{c2} \frac{1}{\sqrt{t}}\right)
\end{aligned} \tag{A.12}$$

After some rearrangement, this leads to

$$\begin{aligned}
C_{\text{loss,cyc,SPM}} \leq & \zeta_{c1} f(\text{DOD} + \zeta_{\theta,n,2})^2 \cdot f(\varnothing V - U_s) \\
& \cdot \exp\left(-\frac{\zeta_1}{\varnothing T_{\text{int}}}\right) \cdot (t - \zeta_{c2} \sqrt{t}) \\
& + \zeta_{c10} f(\text{DOD} + \zeta_{\theta,n,2})^2 \cdot f(\varnothing V - U_s) \\
& \cdot \exp\left(-\frac{\zeta_1}{\varnothing T_{\text{int}}}\right) \cdot Q_{\text{Ah}} \left(1 - \zeta_{c2} \frac{1}{\sqrt{t}}\right).
\end{aligned} \tag{A.13}$$

Appendix B. Abbreviations

DOD	Depth of discharge
EEM	Electrical equivalent model
HEV	Hybrid electric vehicle
LAM	Loss of active material
LLI	Loss of lithium inventory
LMO	LiMn ₂ O ₄
ODE	Ordinary differential equation
OCV	Open circuit voltage
PDE	Partial differential equation
RMSE	Root mean squared error
SEI	Solid electrolyte interface layer
SOC	State of charge
SPM	Single particle model
SPM–EEM	SPM derived EEM

Appendix C. List of symbols

Symbol	Description
$a_{1,2,3,\dots}$	fitting parameters for calendar aging in empirical equations
$b_{1,2,3,\dots}$	fitting parameters for cycle aging in empirical equations
c	cell heat capacity for EEM
c_e	liquid phase (electrolyte) concentration
$c_{s,i}$	solid phase Li-ion concentration
$c_{s,i,\text{max}}$	maximum solid phase Li-ion concentration
	solid phase Li-ion concentration at particle surface
$c_{s,i}^s$	
c_{solv}	solvent concentration
	solvent boundary concentration
c_{solv}^*	
e^-	electron
h_{conf}	convective coefficient
	solvent reduction kinetic aging constant
k_f'	
i	subscript for (n)egative, (p)ositive electrode and separator (sp or sep)
i_s	exchange current density of the side reaction
	exchange current density
i_s^0	
m	cell mass
r	electrode radial variable
t	time
$t_{0,i}$	initial exchange density
t_+	Li-ion transference number
x, y	stoichiometric coefficients
y_n	measured capacity loss for RMSE evaluation of n th aging scenario
	predicted capacity loss
$\hat{y}$	
	mean value
$\bar{y}$	

z	spatial variable in SPM or power factor in empirical equations
A	electrode plate area or estimation parameter for empirical aging model
Q_{Ah}	charge throughput in Ah
C_{loss}	relative capacity loss
C_{rem}	remaining relative capacity
C_n	negative electrode capacity
C_p	cell heat capacity
$C_{1,2}$	capacity of EEM capacitors
$D_{s,i}$	solid phase diffusion coefficient
D_{solv}	solvent diffusion coefficient
DOD	depth of discharge
E_a	activation energy
	process specific activation energy
E_{ak}	
F	Faraday constant
I	current
L	distance between negative and positive current collector
M_{SEI}	SEI molar mass
Q_i	cell electrode's capacity
Q_s	capacity loss due to SEI layer growth
Q_c	charge throughput for SPM based on current I (similar to SOC)
Q	charge throughput for empirical aging
$R_{s,i}$	particle radius
R_{ohm}	ohmic resistance
R_{SEI}	SEI layer total resistance
R	gas constant
$R_{0,1,2}$	resistance of EEM resistors
$Ratio$	ratio between charge sustaining and charge depletion in HEVs
S_n	electroactive surface of the negative electrode
SOC	state of charge
T	temperature in Kelvin
T_{int}	internal cell temperature
T_{ref}	reference cell temperature
T_{amb}	ambient temperature
U	potential
U_i	electrode's half cell OCV
V	voltage
V_{OCV}	open circuit voltage
V_{RC}	voltage loss over RC elements
α	charge transfer coefficient or fitting parameter in empirical equation
$\alpha_{c,r}$	calendar fitting parameter for capacity and resistance empirical aging equation
β	charge transfer coefficient of side reaction
$\beta_{c,r}$	cycle fitting parameter for cap. and res. empirical aging eq.
δ_i	electrode's and separator's thickness
δ_{SEI}	SEI layer thickness
$\epsilon_{s,i}$	active material volume fraction
ϵ_{SEI}	SEI porosity
$\epsilon_{e,i}$	volume fraction of electrolyte
	kinetic overpotential
$\tilde{\eta}_i^k$	
	effective ionic conductivity
κ_i^{eff}	
κ_{SEI}	SEI ionic conductivity
ϕ_e	liquid phase (electrolyte) potential
ϕ_i	negative/positive electrode potential
$\phi_{s,i}$	electrode potential between solid phase of negative/positive electrode
$\phi_{s,i,avg}$	average of $\phi_{s,i}$ during fixed cycle depth
$\phi_{gen,dis}$	generated and dissipated energy
ρ_{SEI}	SEI layer density
	normalized bulk Li-ion concentration
θ_i^b	
	normalized inserted Li-ion concentration at the particle surface
θ_i^s	
	average normalized inserted Li-ion concentration at the particle surface during fixed cycle depth
$\theta_{i,avg}^s$	
$\zeta_{1,2,...}$	constant parameters for simplification of SPM
$\bar{V}$	average voltage

References

- [1] S. Ji, C.R. Cherry, M. Bechle, Y. Wu, J. Marshall, Electric vehicles in China: emissions and health impacts, *Environ. Sci. Technol.* 46 (4) (2011) 2018–2024, <https://doi.org/10.1021/es202347q>.
- [2] J. Weinert, J. Ogden, D. Sperling, A. Burke, The future of electric two-wheelers and electric vehicles in China, *Energy Policy* 36 (7) (2008) 2544–2555, <https://doi.org/10.1016/j.enpol.2008.03.008>.
- [3] M. Weiss, M.K. Patel, M. Junginger, A. Perujo, P. Bonnel, G. van Grootveld, On the

- electrification of road transport – learning rates and price forecasts for hybrid-electric and battery-electric vehicles, *Energy Policy* 48 (2012) 374–393, <https://doi.org/10.1016/j.enpol.2012.05.038> Special Section: Frontiers of Sustainability.
- [4] S.J. Gerssen-Gondelach, A.P. Faaij, Performance of batteries for electric vehicles on short and longer term, *J. Power Sources* 212 (2012) 111–129, <https://doi.org/10.1016/j.jpowsour.2012.03.085>.
 - [5] J. Schmalstieg, S. Kaebitz, M. Ecker, D.U. Sauer, A holistic aging model for Li (NiMnCo)O₂ based 18650 lithium-ion batteries, *J. Power Sources* 257 (2014) 325–334, <https://doi.org/10.1016/j.jpowsour.2014.02.012>.
 - [6] A. Nikolian, J. Jaguemont, J. de Hoog, S. Goutam, N. Omar, P.V.D. Bossche, J.V. Mierlo, Complete cell-level lithium-ion electrical ECM model for different chemistries (NMC, LFP, LTO) and temperatures (–5 °C to 45 °C) – optimized modelling techniques, *Int. J. Electr. Power Energy Syst.* 98 (2018) 133–146, <https://doi.org/10.1016/j.ijepes.2017.11.031>.
 - [7] A. Cordoba-Arenas, S. Onori, Y. Guezennec, G. Rizzoni, Capacity and power fade cycle-life model for plug-in hybrid electric vehicle lithium-ion battery cells containing blended spinel and layered-oxide positive electrodes, *J. Power Sources* 278 (2015) 473–483, <https://doi.org/10.1016/j.jpowsour.2014.12.047>.
 - [8] V. Ramadesigan, P.W.C. Northrop, S. De, S. Santhanagopalan, R.D. Braatz, J.V. Subramanian, Modeling and simulation of lithium-ion batteries from a systems engineering perspective, *J. Electrochem. Soc.* 159 (3) (2012) R31–R45, <https://doi.org/10.1149/2.018203jes>.
 - [9] D.W. Limoge, P.Y. Bi, A.M. Annaswamy, A. Krupadanam, A reduced-order model of a lithium-ion cell using the absolute nodal coordinate formulation approach, *IEEE Trans. Control Syst. Technol.* 26 (3) (2018) 1001–1014, <https://doi.org/10.1109/TCST.2017.2692743>.
 - [10] M. Petit, E. Prada, V. Sauvant-Moynot, Development of an empirical aging model for Li-ion batteries and application to assess the impact of vehicle-to-grid strategies on battery lifetime, *Appl. Energy* 172 (2016) 398–407, <https://doi.org/10.1016/j.apenergy.2016.03.119>.
 - [11] M. Safari, M. Morcrette, A. Teyssot, C. Delacourt, Multimodal physics-based aging model for life prediction of Li-ion batteries, *J. Electrochem. Soc.* 156 (3) (2009) A145–A153, <https://doi.org/10.1149/1.3043429>.
 - [12] J. Marcicki, A. Bartlett, M. Canova, A.T. Conlisk, G. Rizzoni, Y. Guezennec, X.G. Yang, T. Miller, Characterization of cycle-life aging in automotive lithium-ion pouch cells, *ECS Trans.* 50 (26) (2013) 235–247, <https://doi.org/10.1149/05026.0235ecst>.
 - [13] L. Cai, R.E. White, An efficient electrochemical–thermal model for a lithium-ion cell by using the proper orthogonal decomposition method, *J. Electrochem. Soc.* 157 (11) (2010) A1188–A1195, <https://doi.org/10.1149/1.3486082>.
 - [14] A. Bizeray, S. Zhao, S. Duncan, D. Howey, Lithium-ion battery thermal–electrochemical model-based state estimation using orthogonal collocation and a modified extended Kalman filter, *J. Power Sources* 296 (2015) 400–412, <https://doi.org/10.1016/j.jpowsour.2015.07.019>.
 - [15] M.R. Palacin, Understanding ageing in Li-ion batteries: a chemical issue, *Chem. Soc. Rev.* 47 (2018) 4924–4933, <https://doi.org/10.1039/C7CS00889A>.
 - [16] P. Arora, R.E. White, M. Doyle, Capacity fade mechanisms and side reactions in lithium-ion batteries, *J. Electrochem. Soc.* 145 (10) (1998) 3647–3667, <https://doi.org/10.1149/1.1838857>.
 - [17] M. Brüssely, S. Herreyre, P. Biensan, P. Kasztejna, K. Nechev, R. Staniewicz, Aging mechanism in Li ion cells and calendar life predictions, *J. Power Sources* 97–98 (2001) 13–21, [https://doi.org/10.1016/S0378-7753\(01\)00722-4](https://doi.org/10.1016/S0378-7753(01)00722-4).
 - [18] M. Brüssely, P. Biensan, F. Bonhomme, P. Blanchard, S. Herreyre, K. Nechev, R. Staniewicz, Main aging mechanisms in Li ion batteries, *J. Power Sources* 146 (1) (2005) 90–96, <https://doi.org/10.1016/j.jpowsour.2005.03.172>.
 - [19] J. Vetter, M. Winter, M. Wohlfahrt-Mehrens, Secondary batteries — lithium rechargeable systems — lithium-ion — aging mechanisms, in: J. Garche (Ed.), *Encyclopedia of Electrochemical Power Sources*, Elsevier, Amsterdam, 2009, pp. 393–403, <https://doi.org/10.1016/B978-0-444-52745-5.00922-9>.
 - [20] D. Aurbach, Review of selected electrode–solution interactions which determine the performance of Li and Li ion batteries, *J. Power Sources* 89 (2) (2000) 206–218, [https://doi.org/10.1016/S0378-7753\(00\)00431-6](https://doi.org/10.1016/S0378-7753(00)00431-6).
 - [21] E. Peled, S. Menkin, Review – SEI: past, present and future, *J. Electrochem. Soc.* 164 (7) (2017) A1703–A1719, <https://doi.org/10.1149/2.1441707jes>.
 - [22] A.J. Smith, J.C. Burns, S. Zhao, D. Xiong, J.R. Dahn, A high precision coulometry study of the sei growth in Li/graphite cells, *J. Electrochem. Soc.* 158 (5) (2011) A447–A452, <https://doi.org/10.1149/1.3557892>.
 - [23] J. Vetter, P. Novak, M. Wagner, C. Veit, K.-C. Moeller, J. Besenhard, M. Winter, M. Wohlfahrt-Mehrens, C. Vogler, A. Hammouche, Ageing mechanisms in lithium-ion batteries, *J. Power Sources* 147 (1) (2005) 269–281, <https://doi.org/10.1016/j.jpowsour.2005.01.006>.
 - [24] A. Perea, A. Paoletta, J. Dube, D. Champagne, A. Mauger, K. Zaghib, State of charge influence on thermal reactions and abuse tests in commercial lithium-ion cells, *J. Power Sources* 399 (2018) 392–397, <https://doi.org/10.1016/j.jpowsour.2018.07.112>.
 - [25] W. Song, M. Chen, F. Bai, S. Lin, Y. Chen, Z. Feng, Non-uniform effect on the thermal/aging performance of lithium-ion pouch battery, *Appl. Therm. Eng.* 128 (2018) 1165–1174, <https://doi.org/10.1016/j.applthermaleng.2017.09.090>.
 - [26] S.F. Schuster, T. Bach, E. Fleder, J. Mueller, M. Brand, G. Sextl, A. Jossen, Nonlinear aging characteristics of lithium-ion cells under different operational conditions, *J. Energy Storage* 1 (2015) 44–53, <https://doi.org/10.1016/j.est.2015.05.003>.
 - [27] X.-G. Yang, Y. Leng, G. Zhang, S. Ge, C.-Y. Wang, Modeling of lithium plating induced aging of lithium-ion batteries: transition from linear to nonlinear aging, *J. Power Sources* 360 (2017) 28–40, <https://doi.org/10.1016/j.jpowsour.2017.05.110>.
 - [28] E. Peled, The electrochemical behavior of alkali and alkaline earth metals in non-aqueous battery systems – the solid electrolyte interphase model, *J. Electrochem. Soc.* 126 (12) (1979) 2047–2051, <https://doi.org/10.1149/1.2128859>.
 - [29] M.B. Pinson, M.Z. Bazant, Theory of SEI formation in rechargeable batteries: capacity fade, accelerated aging and lifetime prediction, *J. Electrochem. Soc.* 160 (2) (2013) A243–A250, <https://doi.org/10.1149/2.044302jes>.
 - [30] E. Prada, D. Di Domenico, Y. Creff, J. Bernard, V. Sauvant-Moynot, F. Huet, A simplified electrochemical and thermal aging model of LiFePO₄-graphite Li-ion batteries: power and capacity fade simulations, *J. Electrochem. Soc.* 160 (4) (2013) A616–A628, <https://doi.org/10.1149/2.053304jes>.
 - [31] Z. Mao, *Mathematical model and calendar aging study of commercial blended-cathode Li-ion batteries* (Ph.D. thesis), UWspace, 2016.
 - [32] D. Di Domenico, A. Stefanopoulou, G. Fiengo, Lithium-ion battery state of charge and critical surface charge estimation using an electrochemical model-based extended Kalman filter, *J. Dyn. Syst. Meas. Control* 132 (6) (2010), <https://doi.org/10.1115/1.4002475> 061302–061302-11.
 - [33] I. Bloom, B. Cole, J. Sohn, S. Jones, E. Polzin, V. Battaglia, G. Henriksen, C. Motloch, R. Richardson, T. Unkelhaeuser, D. Ingersoll, H. Case, An accelerated calendar and cycle life study of Li-ion cells, *J. Power Sources* 101 (2) (2001) 238–247, [https://doi.org/10.1016/S0378-7753\(01\)00783-2](https://doi.org/10.1016/S0378-7753(01)00783-2).
 - [34] J. Wang, P. Liu, J. Hicks-Garner, E. Sherman, S. Soukiazian, M. Verbrugge, H. Tataria, J. Musser, P. Finamore, Cycle-life model for graphite-LiFePO₄ cells, *J. Power Sources* 196 (8) (2011) 3942–3948, <https://doi.org/10.1016/j.jpowsour.2010.11.134>.
 - [35] X. Han, M. Ouyang, L. Lu, J. Li, A comparative study of commercial lithium ion battery cycle life in electric vehicle: capacity loss estimation, *J. Power Sources* 268 (2014) 658–669, <https://doi.org/10.1016/j.jpowsour.2014.06.111>.
 - [36] N. Omar, M.A. Monem, Y. Firooz, J. Salminen, J. Smekens, O. Hegazy, H. Gualous, G. Mulder, P. Van den Bossche, T. Coosemans, J. Van Mierlo, Lithium iron phosphate based battery – assessment of the aging parameters and development of cycle life model, *Appl. Energy* 113 (2014) 1575–1585, <https://doi.org/10.1016/j.apenergy.2013.09.003>.
 - [37] D. Stroe, M. Ȃwierzcyński, A. Stan, R. Teodorescu, S.J. Andreasen, Accelerated lifetime testing methodology for lifetime estimation of lithium-ion batteries used in augmented wind power plants, *IEEE Trans. Ind. Appl.* 50 (6) (2014) 4006–4017, <https://doi.org/10.1109/TIA.2014.2321028>.
 - [38] E. Prada, D. Di Domenico, Y. Creff, J. Bernard, V. Sauvant-Moynot, F. Huet, Simplified electrochemical and thermal model of LiFePO₄-graphite Li-ion batteries for fast charge applications, *J. Electrochem. Soc.* 159 (9) (2012) A1508–A1519, <https://doi.org/10.1149/2.064209jes>.
 - [39] S.K. Rechkemmer, B. Czech, X. Zang, W. Zhang, O. Sawodny, Adaptability of Li-ion single particle model for lifetime simulation using LFP and LMO cells, 2018 IEEE Conference on Control Technology and Applications (CCTA) (2018) 1415–1420, <https://doi.org/10.1109/CCTA.2018.8511388>.
 - [40] H.J. Ploehn, P. Ramadass, R.E. White, Solvent diffusion model for aging of lithium-ion battery cells, *J. Electrochem. Soc.* 151 (3) (2004) A456–A462, <https://doi.org/10.1149/1.1644601>.
 - [41] J.E.B. Randles, Kinetics of rapid electrode reactions, *Discuss. Faraday Soc.* 1 (1947) 11–19, <https://doi.org/10.1039/DF9470100011>.
 - [42] S.K. Rechkemmer, W. Zhang, O. Sawodny, Modeling of a permanent magnet synchronous motor of an e-scooter for simulation with battery aging model, *IFAC-PapersOnLine* 50 (1) (2017) 4769–4774, <https://doi.org/10.1016/j.ifacol.2017.08.956>.
 - [43] P. Albertus, J. Coutts, V. Srinivasan, J. Newman, II. A combined model for determining capacity usage and battery size for hybrid and plug-in hybrid electric vehicles, *J. Power Sources* 183 (2) (2008) 771–782, <https://doi.org/10.1016/j.jpowsour.2008.05.012>.
 - [44] W.A. Appiah, J. Park, S. Byun, M.-H. Ryou, Y.M. Lee, A mathematical model for cyclic aging of spinel LiMn₂O₄/graphite lithium-ion cells, *J. Electrochem. Soc.* 163 (13) (2016) A2757–A2767, <https://doi.org/10.1149/2.1061613jes>.
 - [45] P. Arora, M. Doyle, A.S. Gozdz, R.E. White, J. Newman, Comparison between computer simulations and experimental data for high-rate discharges of plastic lithium-ion batteries, *J. Power Sources* 88 (2) (2000) 219–231, [https://doi.org/10.1016/S0378-7753\(99\)00527-3](https://doi.org/10.1016/S0378-7753(99)00527-3).